Optimizing Web Testing for Inflectra: Ensuring Quality with Every Click

Test Plan

Website Link: https://www.inflectra.com/

1. Test Strategy

- *Testing Types that will be submitted:*
  - *Functional Testing:* UI, API, Database testing.
  - *Performance Testing:* Load and stress testing using JMeter.
  - *Regression Testing:* Ensuring new features don't break existing ones.
  - *Automation Testing:* Using Selenium for UI and Postman for API.
  - *Security Testing:* If required by the project.

- *Testing Tools:*
  - *Selenium-* UI Automation
  - *Postman/Insomnia-* API Testing
  - *JMeter-* Performance Testing
  - *MySQL-* Database Validation
  - *Cucumber (BDD)-* Behaviour Driven Development
  - *Jenkins-* CI/CD
  - *Git/GitHub-* version control and collaboration

- *Test Execution:*
  - *Manual Tests:* Exploratory testing, UI verification, and basic API checks.

- *Automated Tests:* For sections requiring repetitive testing (UI, API, and Regression).
- *CI/CD Pipeline:* Jenkins will automatically trigger tests after every commit.

2. Scope of Testing

- *In Scope:*
  - *UI Functionality:* Testing navigation, buttons, forms, and links.
  - *API:* Response codes, payload validation, and error handling.
  - *Database:* Validation of data integrity using MySQL queries.
  - *Performance:* Load testing to ensure the application can handle traffic efficiently.

- *Out of Scope:*
  - *Non-Functional Testing:* Security, usability, etc. (unless specified).
  - *Load Testing:* Handled separately through performance testing.

3. Test Objectives

- *Validate Functional Components:* Ensure that all features meet user requirements.
- *Ensure Application Performance:* Ensure the app handles expected traffic loads.
- *Defect Detection:* Identify and report defects in UI, API, or Database functionality.
- *Test Framework Development:* Ensure the testing framework is robust, scalable, and reusable.

4. Test Deliverables

- *Test Cases:* Both manual and automated.
- *Test Execution Reports*: Results of the executed test cases.
- *Defect Reports:* Any issues discovered during testing.

- *Test Summary Reports:* A comprehensive overview of the testing process and results.

5. Risk Management

- *Risk 1: Frequent website changes affecting test case stability.*
  - *Mitigation:* Use dynamic locators and regular updates from the development team.

- *Risk 2: Limited access to the database for validation.*
  - *Mitigation:* Simulate data validation using mock data and API responses.

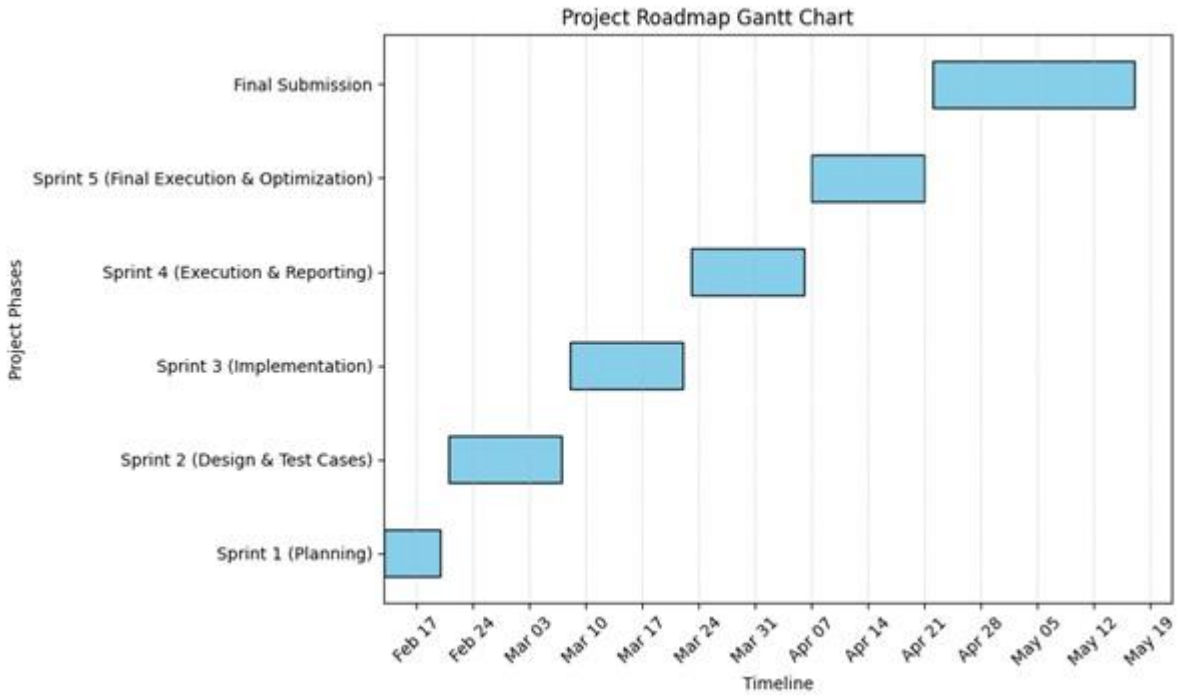
- *Risk 3: Time constraints leading to incomplete test coverage.*
  - *Mitigation:* Prioritize high-risk functionality and automate as much as possible.

6. Entry and Exit Criteria

- *Entry Criteria:*
  - Test environment must be fully set up (tools and servers).
  - The website must be stable and accessible.
  - Test data must be prepared.
  - Test Plan must be approved.

- *Exit Criteria:*
  - All tests (manual and automated) must be executed.
  - Critical/high defects must be resolved.
  - Performance requirements must be met (response time < 4 seconds).
  - The CI/CD pipeline must be integrated and running.
  - A final report must be generated.

7. Project Timeline

Sprint	Duration	Start Date	End Date	Key Milestones
Sprint 1	1 Week	13th Feb	20th Feb	Test Plan Document finalized
Sprint 2	2 Weeks	21st Feb	5th Mar	Test Case Design (Manual & Automated)
Sprint 3	2 Weeks	6th Mar	19th Mar	Framework Implementation (CI/CD, GitHub, Postman)
Sprint 4	2 Weeks	20th Mar	2nd Apr	Test Execution & Reporting
Sprint 5	2 Weeks	3rd Apr	16th Apr	Final Test Suite Execution & Optimization
Final Submission	----------	17th Apr	12th May	Final Project Documentation & Full Report

8. Gantt Chart